

# Deep Dive: 11. Functions

## Concept Summary

Functions are reusable blocks of code that only run when called. They take inputs (arguments) and return outputs.

## Coding Structure

```
def function_name(parameters):  
    # logic  
    return result
```

## Example Code

```
def add_numbers(a, b=5):  
    return a + b  
  
result = add_numbers(10)  
print(result)
```

## Code Explanation

- Line 1: Defines a function. 'a' is a required parameter. 'b' is an optional parameter with a default value of 5.
- Line 2: The 'return' keyword sends the calculated result back to the caller.
- Line 4: Calls the function, passing 10 for 'a'. Since 'b' is not provided, it defaults to 5. Result is 15.

## Practice Problems

1. Write a function that takes a string and returns it in all uppercase.
2. Write a function that calculates the area of a circle, requiring a radius parameter and defaulting pi to 3.14.